River Delta Processes in Earth System Models

- A New Bifurcation Scheme

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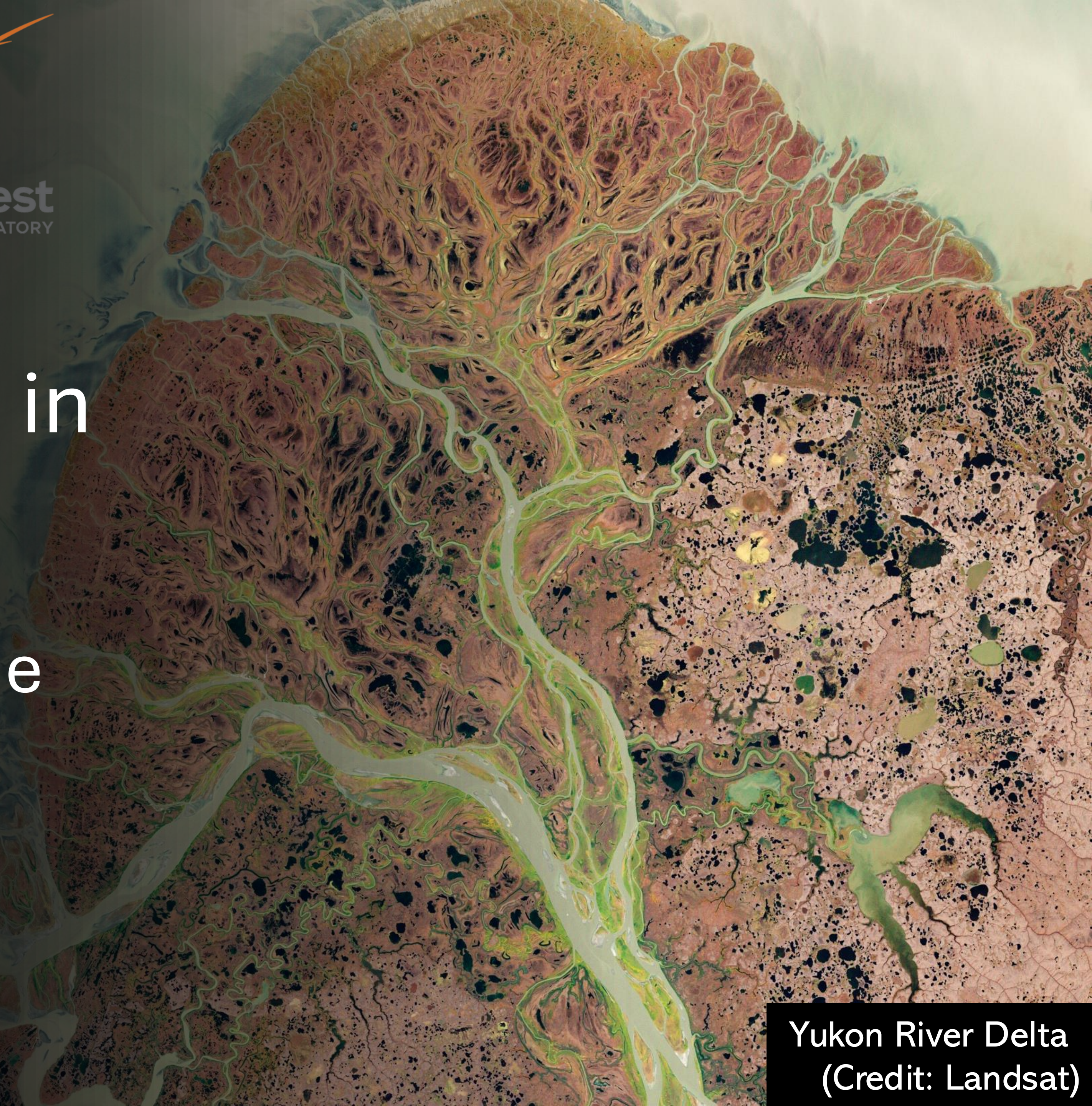
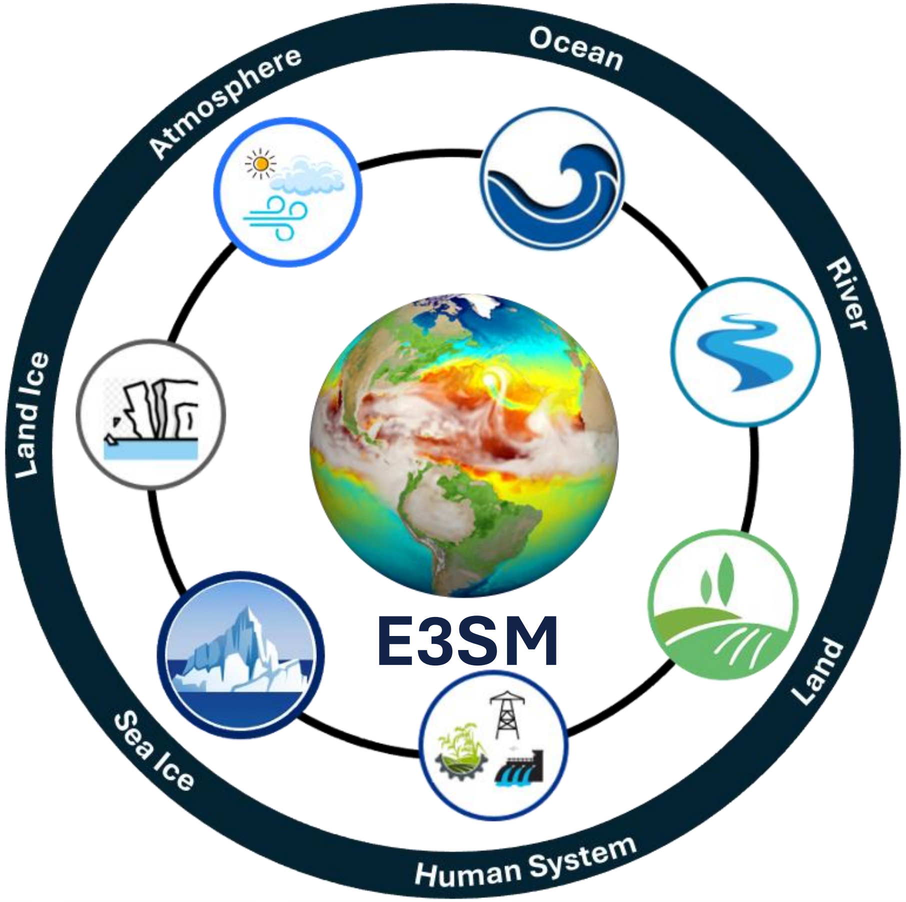


Yukon River Delta
(Credit: Landsat)

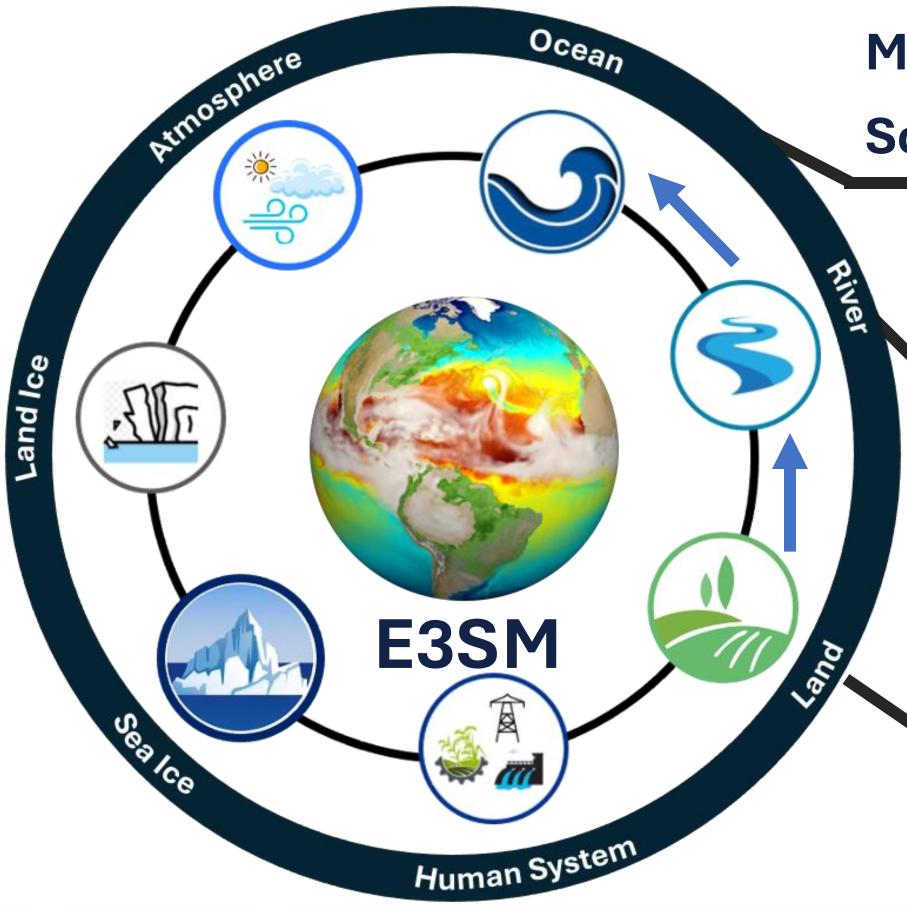
Energy Exascale Earth System Model (E3SM)



US DOE's E3SM is a state-of-the-science Earth system model development and simulation project to investigate energy-relevant science using code optimized for DOE's advanced computers

(e3sm.org)

Energy Exascale Earth System Model (E3SM)

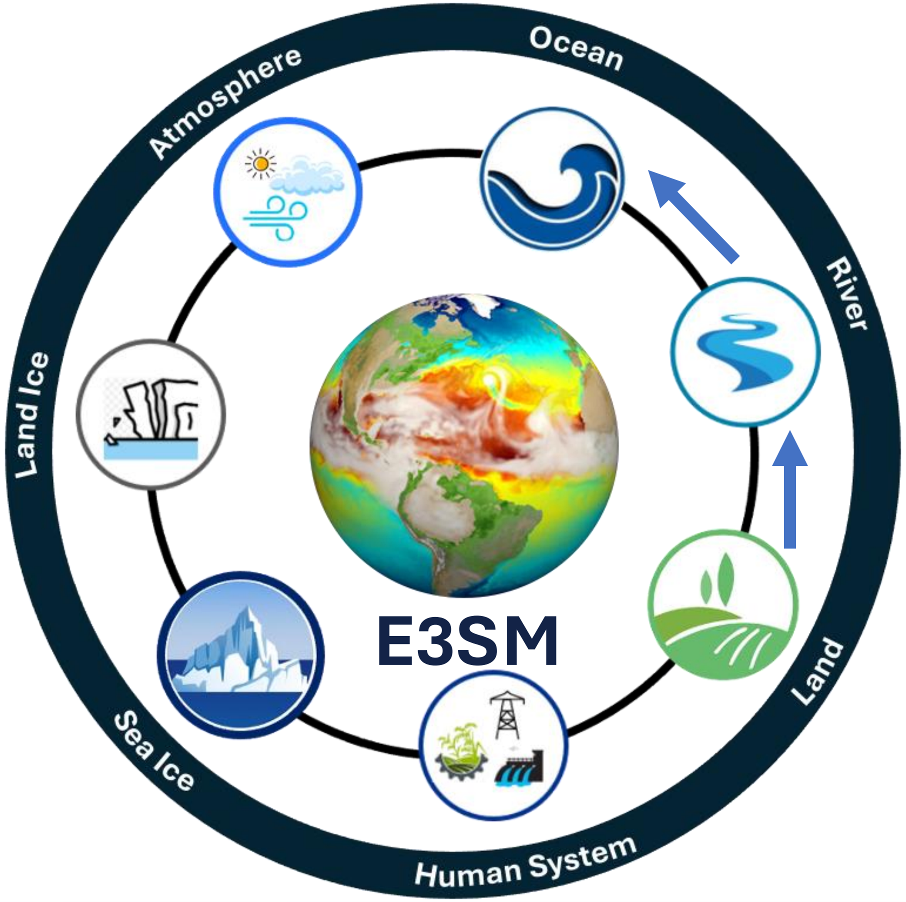


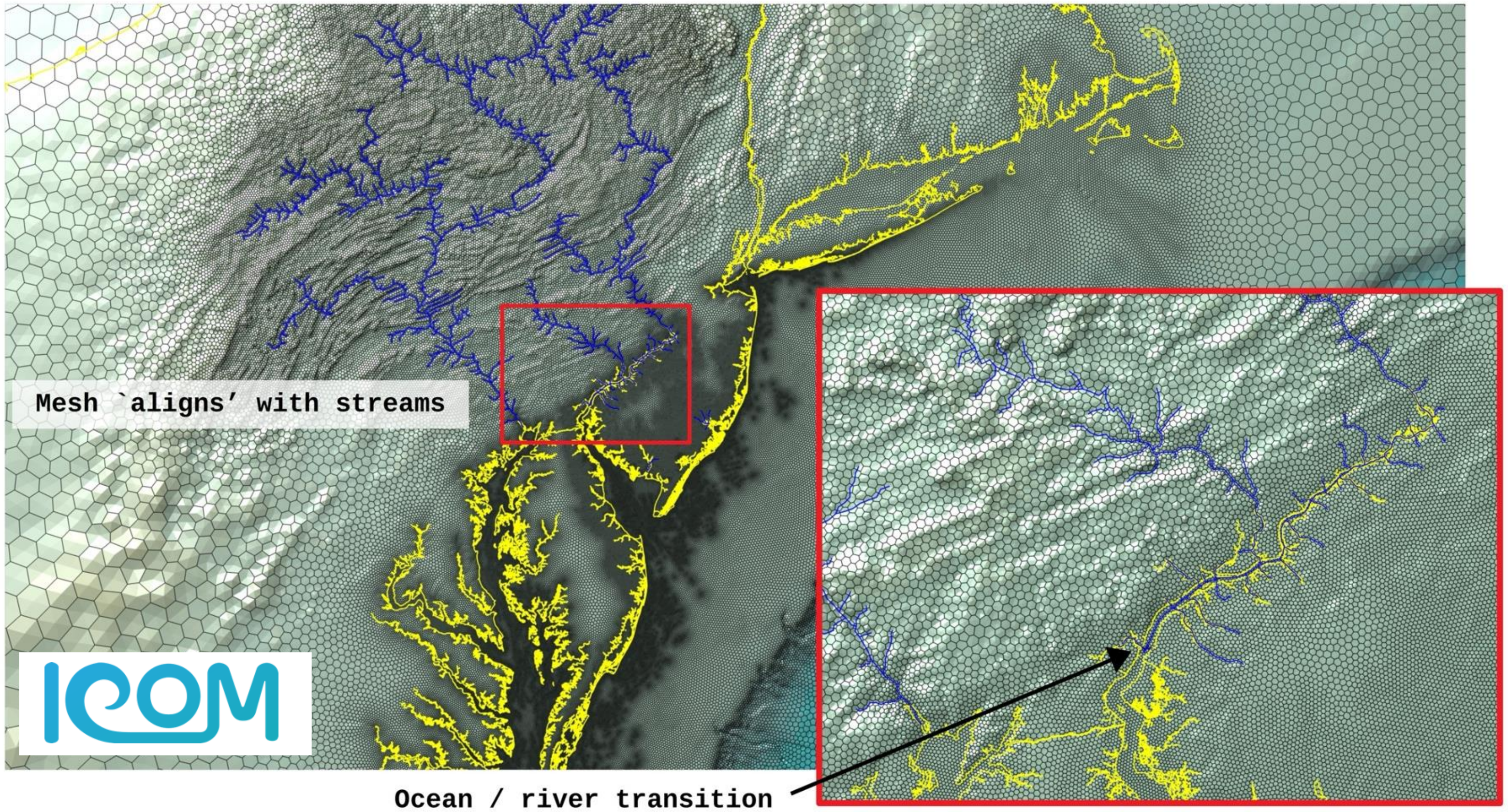
Model for Prediction Across Scales – Ocean (MPAS-Ocean)

Model for Scale Adaptive River Transport (MOSART)

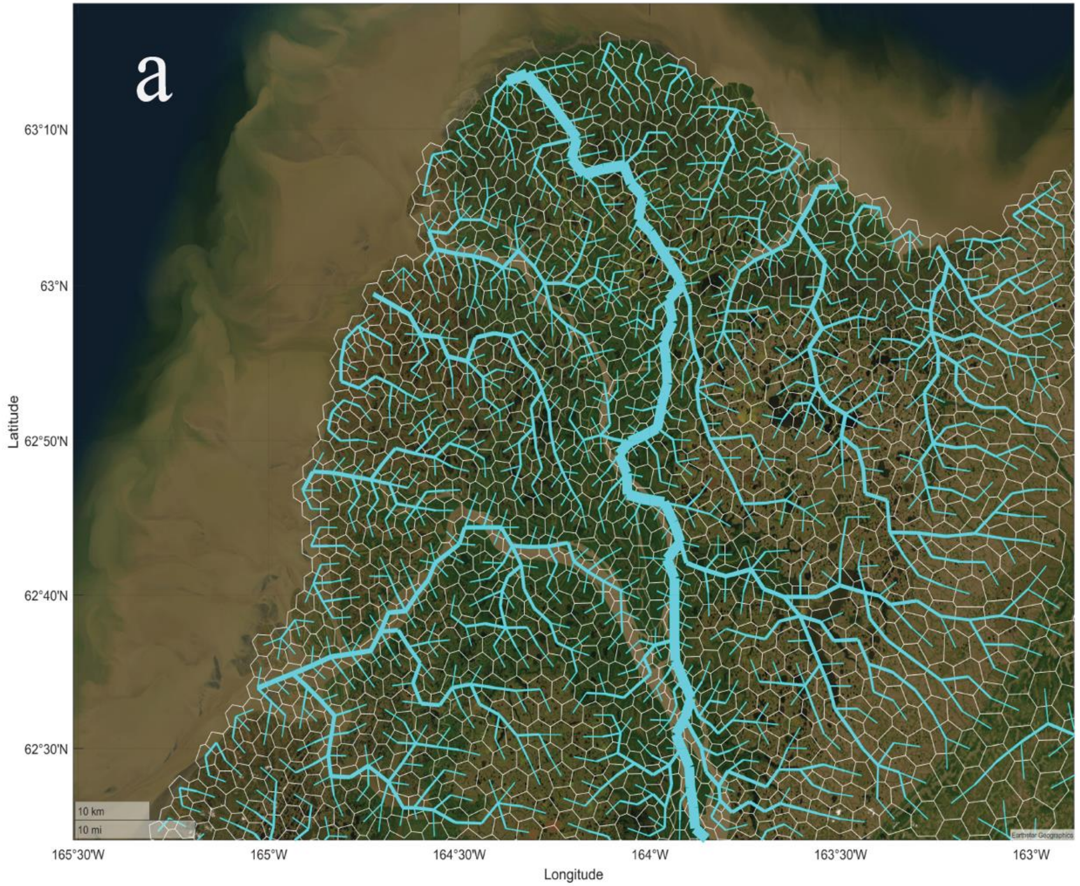
E3SM Land Model (ELM)

MPAS Seamlessly Connects Land, River, and Ocean





Challenge – The "Single Channel" Limitation



- Unrealistic freshwater input to the ocean
- Numerical instabilities at the coast
- Inaccurate sediment plumes

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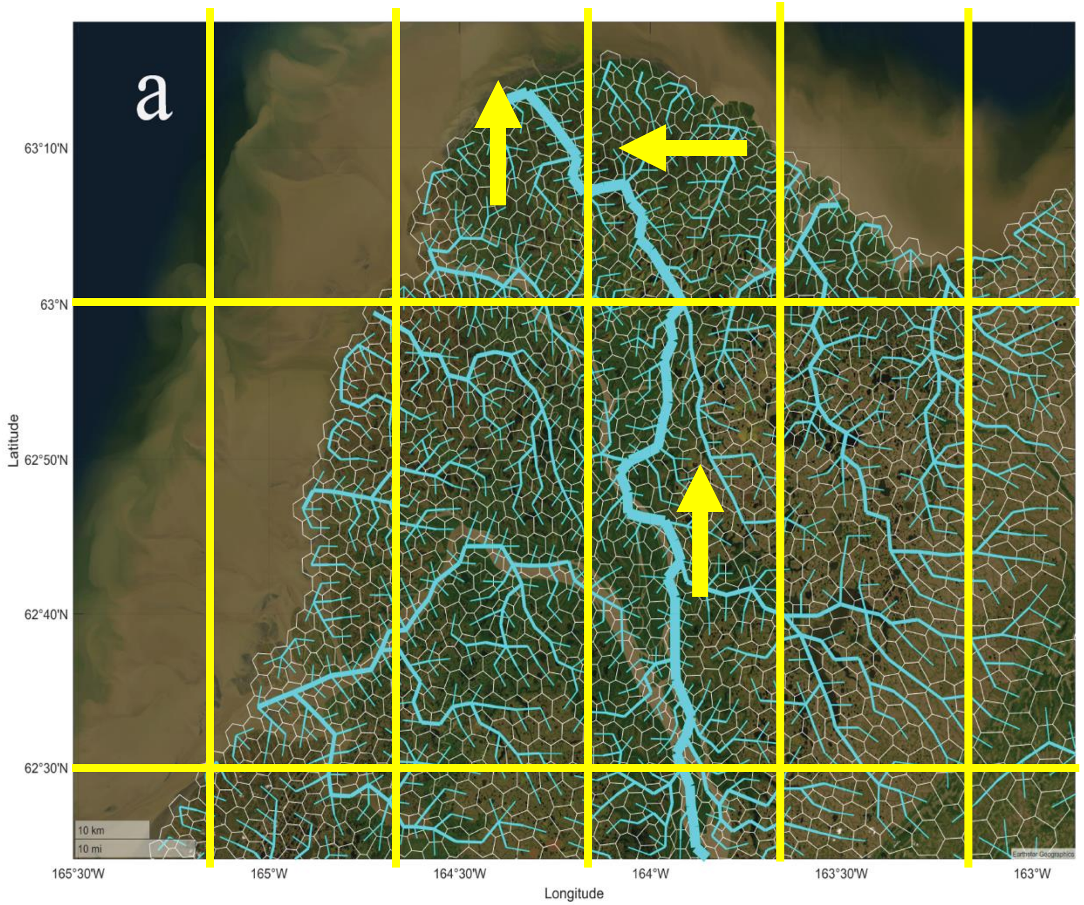

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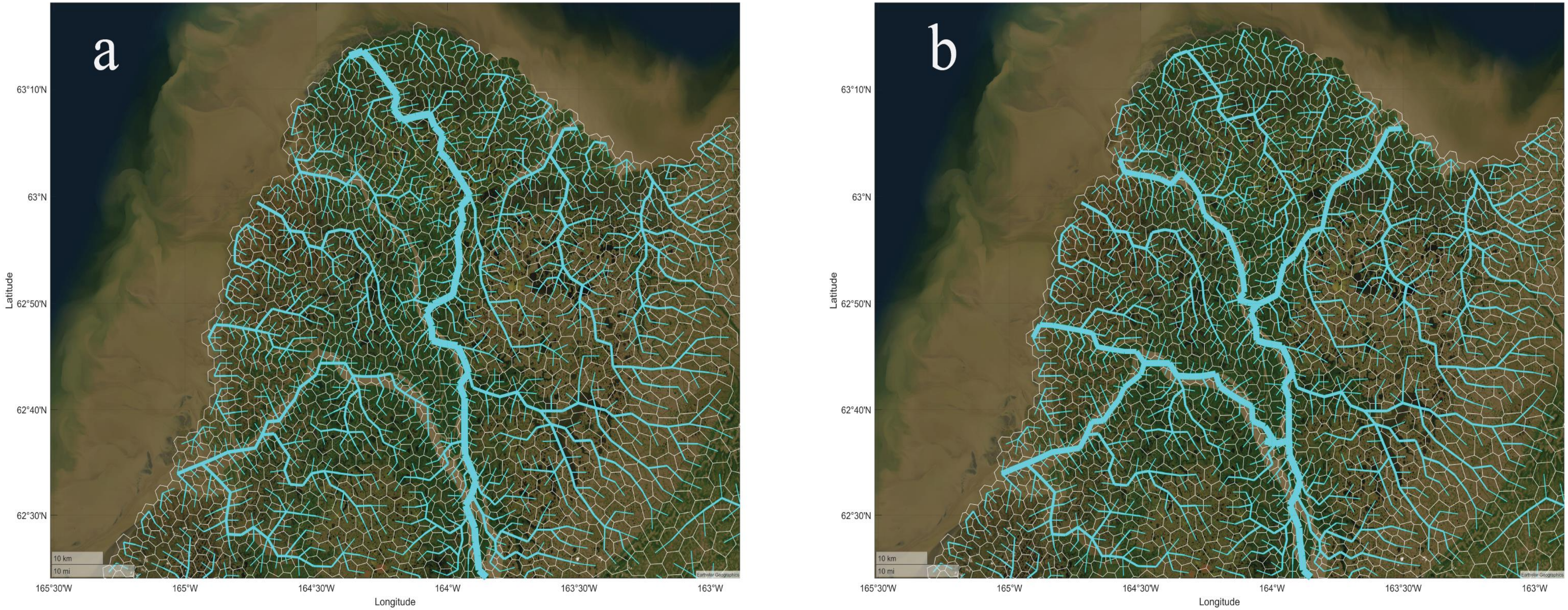
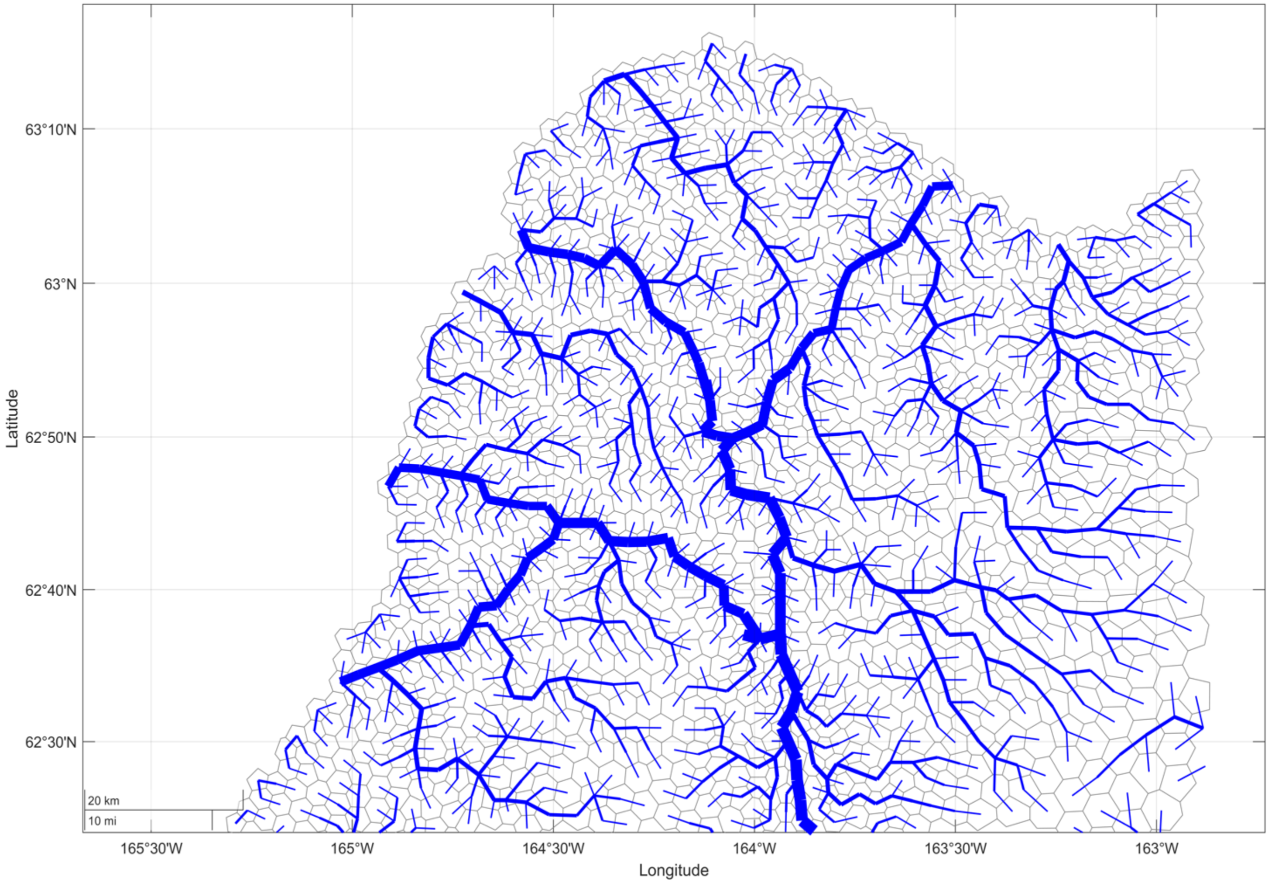
Solution – A New Bifurcation Scheme



Solution – A New Bifurcation Scheme



- **Flexible Splitting:** Allows user-definable bifurcation points and flow-splitting ratios.
- **Grid Agnostic:** Works on both Lat-Lon and unified MPAS meshes.
- **Automation:** Python tools generate the channel network automatically.

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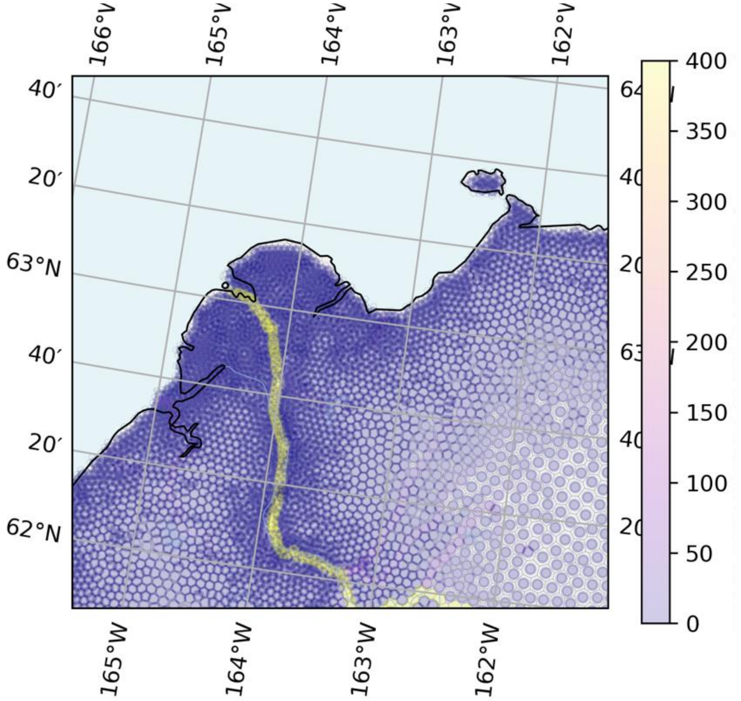
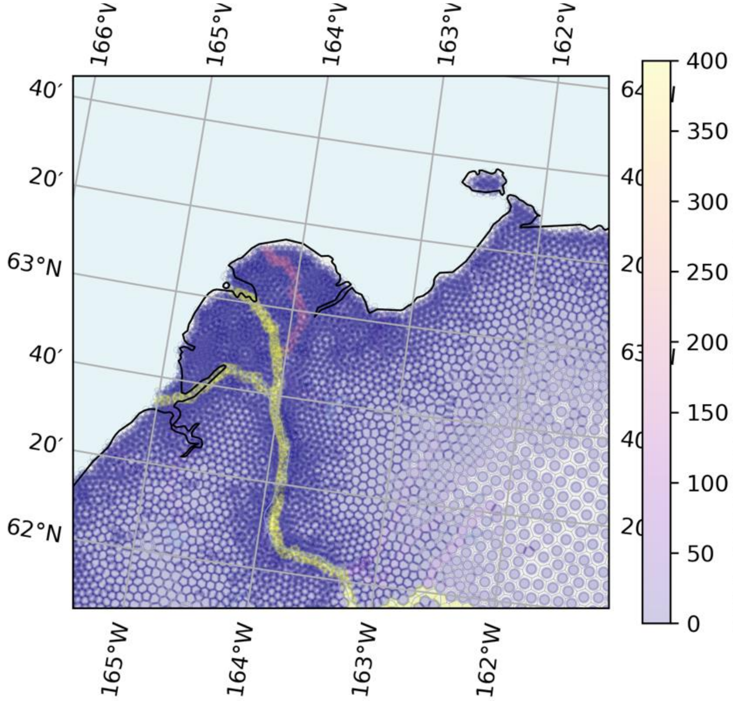
Yukon River Delta Test Case

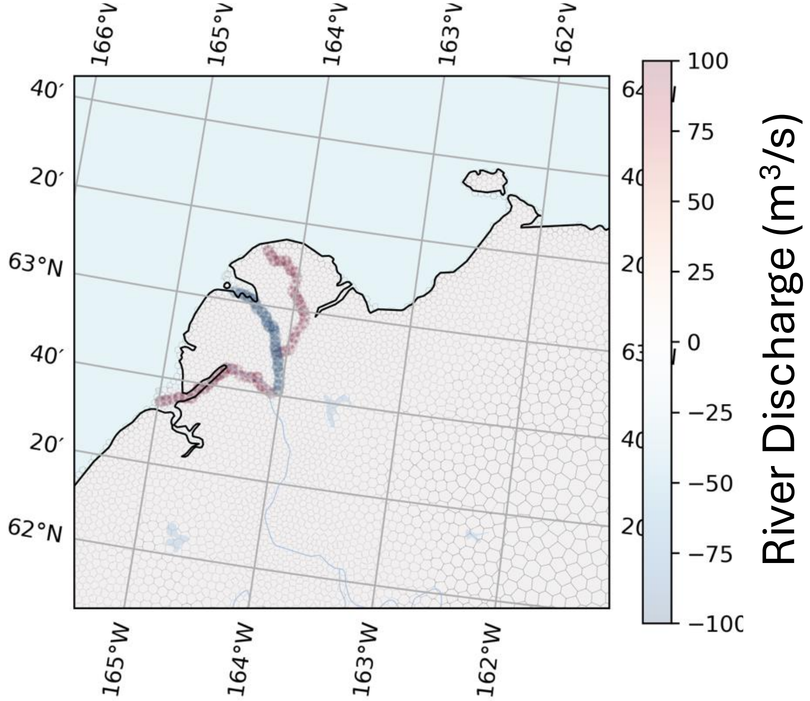
Baseline (Single Channel)



Bifurcation



Bifurcation - Baseline



Future Outlook

- **Immediate Goals:** Validate the simulated streamflow and enable sediment transport through the new channel networks.

- **Expanding the Scope:** Extend the application from the current testbeds to the full pan-Arctic domain to assess broader impacts.

- **Long-term Vision:** Develop a machine learning approach to create dynamic flow-partitioning ratios from remote sensing data, allowing the river network to evolve over time.

- E3SM V1-V3 (coupled simulation)
  - MOSART (natural river process) (*Li et. al. 2013*)
- E3SM V2/V3 (features integrated into E3SM for BGC simulation campaign)
  - MOSART-WM (water management) (*Voisin et al. 2013*)
  - MOSART-Inundation (two schemes) (*Luo et al. 2017*)
  - MOSART-ELM two-way irrigation coupling (*Zhou et al. 2020*)
- Other new features **turned off** in coupled mode
  - MOSART-Heat (water temperature) (*Li et al. 2015*)
  - MOSART-Sediment (sediment transport) (*Li et al. 2021*)
- Ecosystem projects supported MOSART (ongoing) features
  - Routing with unstructured (MPAS) mesh (ICoM) (*Liao et al. 2023*)
  - MOSART-ELM two-way inundation coupling (ICoM) (*Xu et al. 2022*)
  - Two-way coupling between river and ocean (ICoM) (*Feng et al. 2022*)
  - Inter-basin water transfer (ICoM) (*Zhou et al. in prep*)
  - MOSART-Urban (ICoM) (*Li et al. in prep)*
  - MOSART-WM-Hydropower (offline) module (IM3, 9505) (*Zhou et al. 2018*)
  - MOSART-ATS coupling in Arctic basins (InteRFACE) (*Gao et al. in prep*)
  - MOSART-Bifurcation (InteRFACE) (*Zhou et al. in prep*)
  - MOSART-GCAM water coupling (E3SM) (*Zhou et al. in prep*)
  - MOSART-WM ML driven dam operation (ICoM) (*Hoang et al. 2025)*

Thanks for your attention!

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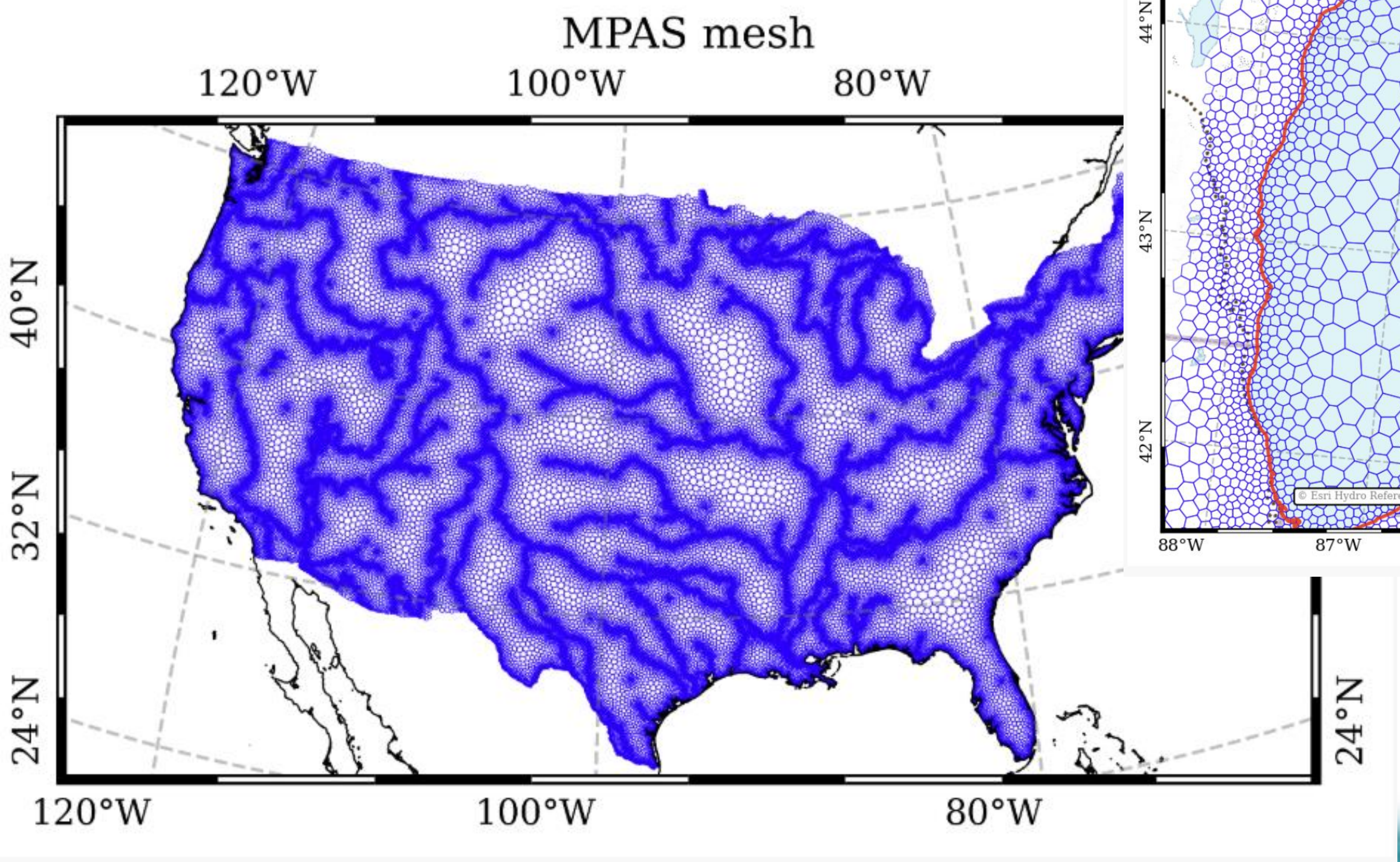
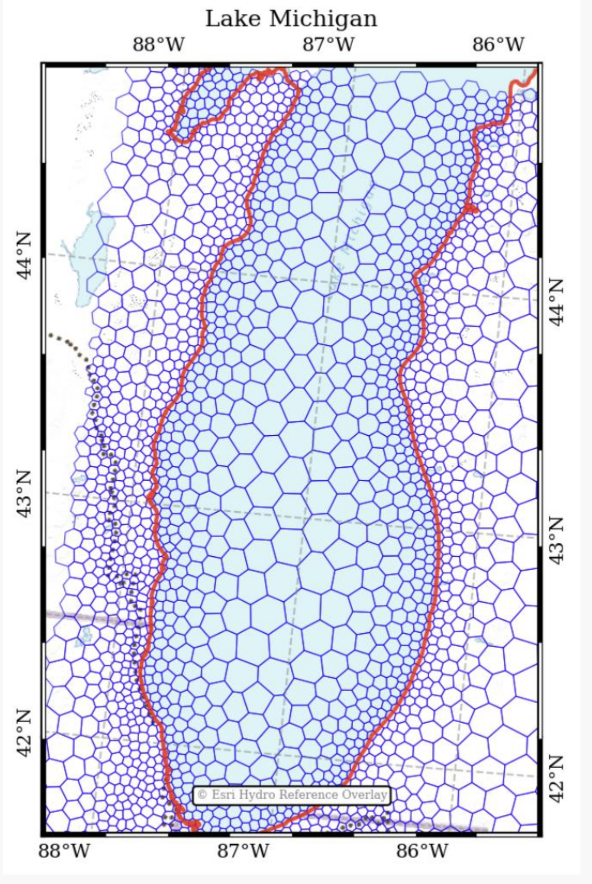
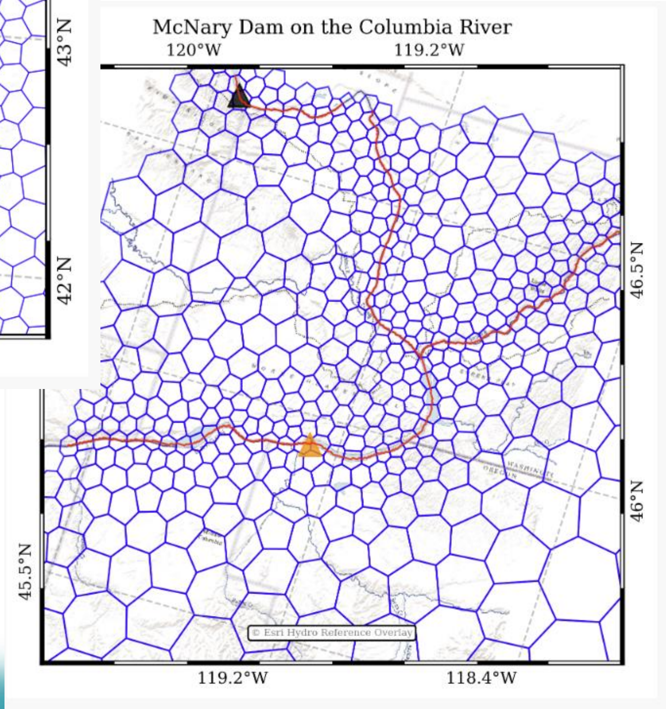
https://simhydro.com

Backup


MPAS mesh
120°W
100°W
80°W
40°N
32°N
24°N
24°N
120°W
100°W
80°W


Lake Michigan
88°W
87°W
86°W
44°N
43°N
42°N
44°N
43°N
42°N
© Esri Hydro Reference Overlay
88°W
87°W
86°W


McNary Dam on the Columbia River
120°W
119.2°W
46.5°N
46°N
45.5°N
© Esri Hydro Reference Overlay
119.2°W
118.4°W

Backup – Inter Basin Water Transfer